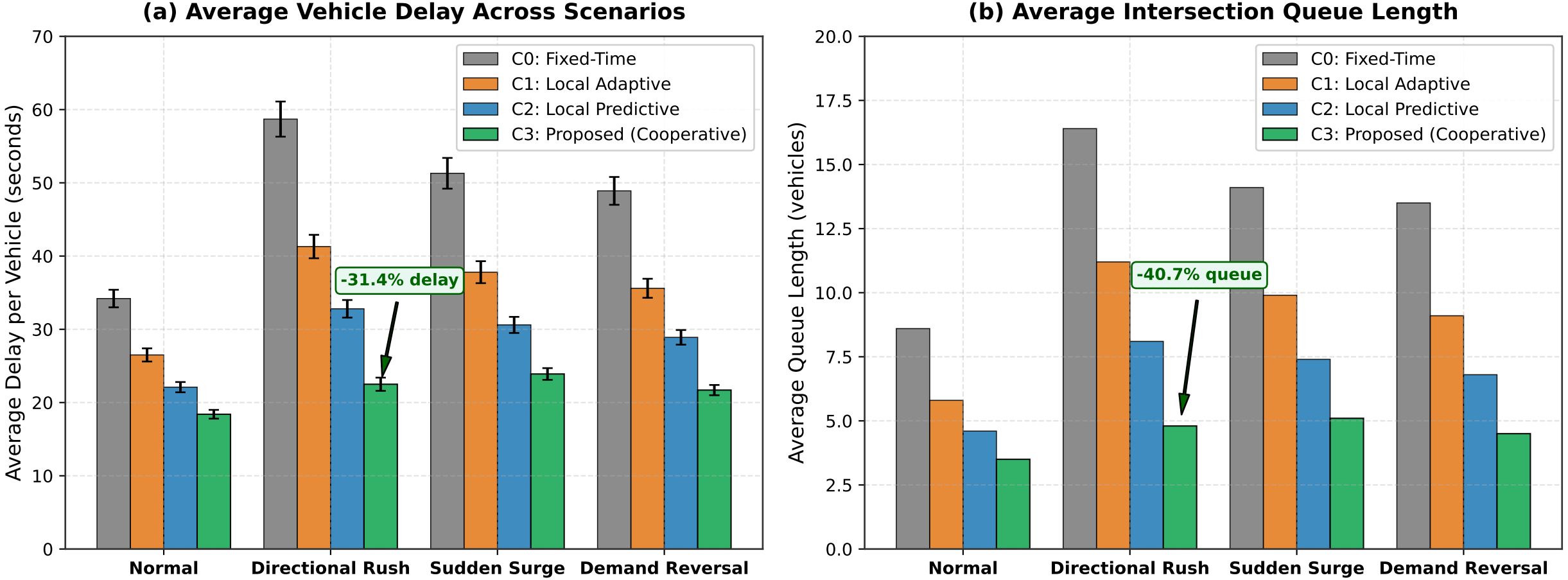
(a) Average Vehicle Delay Across Scenarios
Average Delay per Vehicle (seconds)
C0: Fixed-Time
C1: Local Adaptive
C2: Local Predictive
C3: Proposed (Cooperative)
-31.4% delay
Normal
Directional Rush
Sudden Surge
Demand Reversal
(b) Average Intersection Queue Length
Average Queue Length (vehicles)
-40.7% queue